

Multilingual Speaker Diarization+ASR on Indic YouTube

99 YouTube clips · 12.26 h · nine Indic scripts · 2–8 speakers/clip · 7.13 % overlapped speech. DER at **collar 0.0 with overlap scored**, per the brief. Every figure below is re-derived from committed artefacts and reproduced end-to-end by `main.ipynb`. · Full documentation: `README.md` · lab notebook: `obs.txt`

```
Saaras v3 @ DOVER-Lap(
  community-1, reverb-v2, diarizen-large)
+ per-clip script correction (Sonnet 4.6, T=0)
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metric	baseline	final	Δ
cpWER	0.3957	0.3049	-23.0 %
WDER	0.1128	0.0603	-46.6 %
WER	0.2728	0.2617	-4.1 %
DER	0.2521	0.2442	n.s.
JER	0.3834	0.3608	-5.9 %
speaker-count acc.	60.6 %	79.8 %	+19.2 pt

Baseline is Saaras v3 on the best single diarizer (`reverb-v2`). The final system wins on **92 of 99 clips** individually.

Against the achievable floor. Feeding the reference transcript back as if perfectly recognised, on the reference diarization, still scores cpWER **0.1242** — one transcript cannot carry two simultaneous speakers. The final system sits **0.1807 above that floor**, not 0.3049 above zero.

1 · DATA AND DISCIPLINE

99 of 100 clips extracted; 9 have no overlap. Two properties drive every decision. Overlap is pervasive rather than an edge case. And the reference is **code-switched with a dual-form convention** — 20,599 parenthesised Latin glosses, e.g. `काँफी (coffee)` — so *surface form*, not just meaning, is scored.

Ground truth is never a pipeline input. Stages receive `ClipInput`, a frozen dataclass that structurally carries no reference field. Speaker counts never reach a diarizer; language never reaches a recogniser. The 50/49 dev/test split is frozen in `results/split.json`; test was scored once.

2 · WHAT PRODUCED THE GAIN

step	cpWER	Δ
baseline, Saaras v3 @ <code>reverb-v2</code>	0.3957	—
1 · DOVER-Lap fusion front-end	0.3381	-0.0576
2 · provenance bug fix	0.3181	-0.0200
3 · per-clip script correction	0.3049	-0.0133

2.1 · DOVER-Lap fusion

Hungarian label alignment onto a growing centroid, then per-frame voting with **each speaker thresholded independently** so overlap survives — an argmax vote cannot emit two simultaneous speakers. `pyannote-3.1` is excluded: it shares `segmentation-3.0` with `community-1` and differs by 0.320 s of miss+FA over 12.4 h, so under majority voting they would be one vote counted twice.

GT-free at inference — `dover_lap` reads only the three constituent RTTMs and the clip duration, which comes from the CSV manifest, not from any annotation. The configuration was selected on **dev** and frozen in `results/fusion_config.json` before test evaluation. The operator is GT-free; the selection used dev DER. Those are kept apart deliberately.

2.2 · A provenance bug — the largest lesson

27 of 99 clips had been transcribed against a *stale two-system fusion*, so the measured “fusion” result was partly not the fusion.

`settings_key` existed precisely to catch stale work and missed it, because it fingerprinted **decoding** and not **segmentation**. The fix was a `segmentation_key`: a hash of the turn boundaries, re-derived from each transcript’s stored spans at audit time and compared against the diarization on disk. Worth **-0.0200 cpWER** and **-41 % insertions**, from no modelling change at all — for scale, the GT-perfect ceiling for overlap deduplication was -0.023.

2.3 · Per-clip script correction

The reference writes code-switched English *phonetically in the native script*; Saaras writes Latin. **3,481 of 21,357 substitutions are correctly-recognised words scored wrong for their script**. Sonnet 4.6 rewrites the script of a token and nothing else, seeing only the clip it is correcting — no corpus vocabulary, no other clip, never the reference. Replacement is whole-token, so token count is invariant and words cannot be added, deleted, reordered or translated.

3 · HELD-OUT VALIDATION

Model, prompt, temperature and every guard were frozen and committed at `f284e59` before the test half was scored.

split	before	after	Δ	95 % CI	W/L
dev (50)	0.2676	0.2619	-0.0056	[-.011, -.003]	26/0
test (49)	0.3552	0.3364	-0.0189	[-.024, -.005]	29/0

3,249 of 121,243 tokens changed (2.68 %): **2,232 helpful edits, 0 harmful, 0 cross-script corruptions**. Zero harmful is structural — the reference holds 27 Latin tokens in 123,896, so a Latin hypothesis token is almost never already a match and an edit can only help or be inert. The downside is bounded at zero.

4 · THE DIARIZATION NUMBER, STATED PLAINLY

comparison	Δ DER	95 % CI	W/L
<code>community-1</code> → fusion	-0.0193	[-.025, -.014]	83/16
<code>pyannote-3.1</code> → fusion	-0.0195	[-.031, -.009]	80/19
<code>diarizen-large</code> → fusion	-0.0184	[-.029, -.007]	80/19
<code>reverb-v2</code> → fusion	-0.0079	[-.035, +.016]	42/57

The fusion beats three of four constituents decisively. Against `reverb-v2` the DER difference is **not significant** — the interval spans zero and `reverb-v2` wins 57 of 99 clips. An earlier draft reported “-3.1 % relative DER” as the Step 4 result; that is **withdrawn**. What the fusion does win is JER, confusion, speaker-count accuracy (60.6 % → 79.8 %) and, decisively, the downstream ASR result — cpWER 0.3957 → 0.3181, which is significant.

5 · ERROR BY LANGUAGE

script	clips	WER	cpWER	WDER
Telugu	12	0.3535	0.4123	0.1127
Malayalam	7	0.3322	0.3900	0.0736
Oriya	9	0.3464	0.3868	0.0535
Bengali	8	0.2467	0.3750	0.1233
Gujarati	12	0.2950	0.3332	0.0673
Kannada	9	0.2372	0.3281	0.0696
Gurmukhi	7	0.2770	0.3229	0.0548
Tamil	10	0.2471	0.2723	0.0280

Devanagari	25	0.1906	0.2006	0.0293
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Telugu is 2.06× Devanagari on cpWER — language is a far larger axis of variation than anything else measured. Bengali has the third-best WER but the worst WDER: words recognised well, speakers attributed badly. Groups are 7–25 clips, so read the ordering as indicative.

6 · WHAT DID NOT WORK

intervention	outcome
1-of-3 overlap voting	DER .2442→.3193; FA .050→.218
ConvTasNet separation	word recovery 52.6→50.0 %
LLM contextual refinement	null; all abandon criteria fired
segmentation transplant	confusion worsened; no grid cell won
blind boundary padding	−0.0031 DER — too small to ship
overlap deduplication	helped before the bug fix, hurts after

Two further ideas — an MSDD-style verifier and an OSD-and-constituent intersection — reached only candidate-precision audits (11.5% and 23.6%) and were dropped before implementation; they are reported as **audits, not experiments**. The separation result is scoped narrowly: an *out-of-domain* 8 kHz separator, given *oracle* overlap regions, does not help; a 16 kHz in-domain separator is **untested**.

Whisper was rejected on evidence, across three probes: configuration is not the problem (eight configurations span 0.906–0.964 WER), language ID is not the problem (oracle language moves WER by ≤0.007), and it is an Indic capability gap — IndicConformer-600M scores **WER 0.4158 against Whisper’s best 0.9058**, 2.2× better from a model 2.6× smaller. On 92 of 99 clips Whisper emitted fluent English *translation* rather than transcription, and it has **no Oriya** at all.

7 · WHERE THE ERROR STILL IS

Overlap dominates. It carries 52.5% of all miss and 27.4% of DER error, and overlapped words are deleted at **3.9× the clean rate**. Fusion overlap recall is only 21.5% (678 s of 3,148 s), and 882 s of the 1,560 s it marks is in the wrong place. A perfect separator is worth **−0.031 WER** — the largest ceiling measured here.

The brief’s hint does not hold on this corpus. “Repeated text across a speaker boundary suggesting a false split” occurs **36 times in 6,312 segments** and does not predict error (lift 1.09×). The signals that do work are duration proxies pointing at the wrong targets: segments under 2 s have a 56% error rate but hold only 30% of the wrong-labelled time, while the 36% in long segments has a 5.6% error rate — flagging it would be 94% false positives.

Ground-truth quality is a measurement floor, and the audit is recorded but not fully resolved. 39 of 99 clips carry a reference displaced 1.0–5.0 s, 38 in the same direction. Applying the **23** that met the auto-accept bar is a **diagnostic, not the headline** — every reported metric uses raw ground truth — but it moves fusion DER 0.2442 → 0.1959 and *inverts the ranking among single systems*: the least temporally precise model looked best because the reference was wrong. **The other 16 remain open**, held for manual review because the constituent systems disagreed on the lag. Two further families are catalogued and likewise unfixed: 39 unannotated stretches carrying 218 s of false alarm, and 16 over-long reference turns carrying 563 s of miss — 97% of it one pathological speaker label.

8 · ENGINEERING LESSONS

- **Provenance outranks modelling**. The largest single ASR gain after the front-end swap was a bug fix. “Is the input what I think it is” should be checked before “is the model good enough”.
- **A silent failure that looks like a valid result is worse than a crash**. Twice: a stale fusion that `is_done()` happily skipped, and a truncated LLM response that parsed to `{}` and was counted as 431 abstentions.
- **Aggregates hide inverted rankings**. GT correction moved fusion DER by 19.8% and reversed which single model is best. Any claim resting on a 0.01 DER gap is noise — **including one of our own**, now withdrawn.
- **Structural impossibility beats validation**. Omitting speakers and timestamps from the LLM’s response schema removed a class of failure no prompt instruction would have guaranteed.

9 · LIMITATIONS

- **DER parity with reverb-v2**. The fusion’s case rests on JER, speaker count and downstream cpWER, not on DER.
- **99 clips is small** — 0.01-level DER differences are not resolvable — and the dev/test boundary is stratified on reference-derived fields: held out from tuning, but drawing it was not a GT-blind act.
- **Step 4b is script-only**. It cannot fix a misheard word, and 31 % of its edits are correct transliterations the annotator spelled differently — unrealised headroom.
- **An earlier Step 4b variant** builds one lookup table from all 99 clips’ hypotheses, scoring 0.0032 better (0.3017) but **transductive**: with a dev-built vocabulary its test gain collapses to −0.0042. The per-clip variant ships because its number survives review.
- **Whisper’s 99-clip row is a known-broken configuration**, excluded from the headline; IndicConformer and Saaras v4 cover only 10 and 9 clips.
- **The GT audit is a diagnostic, not a correction pass**. Spot-checks covered 15–20 of the 39 flagged clips and per-clip verdicts were not recorded, so the attestation is at that granularity. Headline numbers use raw ground truth throughout.

10 · REPRODUCING

`main.ipynb` runs Steps 1–5 end to end from committed checkpoints — **no GPU, no API key, no gated model access**; a `SUBSET = 10` toggle verifies the pipeline in under a minute. Verified from a clean clone. Per-video tables `step2_metrics.csv` (495 rows) and `step3_metrics.csv` (638 rows) pool exactly to every figure above.

References. Raj+ *DOVER-Lap* SLT 2021 · Bredin *pyannote.audio/..metrics* · Han+ *DiariZen* ICASSP 2025 · Rev.ai *Reverb* v2 · Sarvam AI *Saaras* v3 · Radford+ *Whisper* ICML 2023 · AI4Bharat *IndicConformer-600M* · Watanabe+ *CHiME-6* 2020 (cpWER) · El Shafey+ 2019 (WDER) · Anthropic *Claude Sonnet* 4.6. Full list in *WRITEUP.md*.